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Climate Change and Workers' Health and Safety: An Umbrella Review

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Abstract

Background: *This umbrella review synthesizes current evidence on the health and safety impacts of climate change among workers. In particular, it aims to identify the main workers' categories at the highest risk, the main health and safety outcomes, methodological limitations, and future directions reported in the literature, as well as preventive strategies described in recent reviews.* **Methods:** *A comprehensive search was conducted in PubMed, Scopus, and Web of Science. Systematic and scoping reviews addressing climate-related occupational health and safety outcomes were included. Study selection and data extraction were performed independently by two reviewers. Methodological quality was assessed using the ROBIS tool, and findings were synthesized in accordance with PRISMA and PRIOR guidelines.* **Results:** *Most of the available evidence concerns outdoor occupations, particularly agriculture and construction, where exposure to high temperatures is associated with increased risks of injuries and heat strain. Emerging data also highlight links with mental health, renal disease, and chronic conditions. However, the literature remains fragmented, with few meta-analyses and substantial heterogeneity. Preventive strategies have been proposed, but their implementation remains inconsistent.* **Conclusions:** *Evidence confirms that climate change has become a determinant of occupational health and safety. Future research should strengthen methodological coherence, adopt longitudinal approaches, and address vulnerable populations.*

1. Introduction

1.1. Background

Climate change is a recognized global challenge [1], with consequences for both ecosystems and human health due to (i) the increase in global average temperatures and (ii) the intensification of heat waves, in terms of frequency, intensity, and duration [1-3]. The working population is particularly vulnerable, as workers are often exposed to adverse climatic conditions daily, frequently for prolonged periods, and under working conditions that may increase their exposure compared to the general population [2, 4]. Occupational exposure mostly occurs in agriculture, construction, manufacturing, and transport, where workers commonly operate in settings where climatic and thermal risks cannot be fully controlled. Health consequences can be particularly pronounced in occupations involving heavy physical labor, limited or absent access to climate-controlled environments, or strict productivity targets that reduce opportunities for rest and recovery. The negative impacts of climate change on

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health reported in the scientific literature are many and varied. Among others, reported direct effects on occupational health include heat-related illnesses, dehydration, cardiovascular and renal dysfunction [1, 2, 5], while indirect effects include injuries related to reduced concentration and psychomotor performance [2, 3, 5, 6], as well as mental health problems and an increased risk of vector-borne infectious diseases for outdoor workers, linked to the expansion of vector habitat driven by rising temperatures [1, 5].

1.2. Problem Statement

Despite the increasing number of systematic reviews in recent years focusing on aspects related to occupational exposure to heat and climate change, the scientific literature remains fragmented and heterogeneous. Several challenges persist, including a lack of harmonization of definitions for heat stress and climate change exposures, inconsistencies in exposure assessment methods, and variations in outcome reporting across studies. Moreover, several literature reviews focus exclusively on specific sectors (e.g., the most affected sectors such as agriculture or construction), geographic regions, or specific health outcomes. In addition, variations in study designs, exposure metrics, and their definitions make it difficult to generalize and compare results across studies.

1.3. Aims of the Study

The main aim of this umbrella review is to provide a comprehensive synthesis of existing systematic and scoping reviews on the effects of climate change on occupational health and safety, and to evaluate the effectiveness of preventive interventions. Specific objectives of this work are:

1. To identify groups of workers most exposed to climate-related risks, with specific attention to urban settings.
2. To summarize the main identified health and safety effects of climate change on workers.
3. To highlight the main knowledge gaps and identify priorities for future studies and evaluations.
4. To review suggested strategies to prevent and mitigate these impacts (e.g., organizational measures, technical interventions) and their effectiveness.

2. Methods

The protocol of the present study was registered with PROSPERO (registration number: CRD420251085510). No deviations from the registered protocol were made. This umbrella review was conducted following the PRIOR (Preferred Reporting Items for Overviews of Reviews) statement [7] and in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [8]. The PRISMA flowchart illustrating the selection process is reported in Figure 1.

2.1. Search Strategy

A comprehensive literature search was performed using three electronic databases: PubMed, Scopus, and Web of Science. The initial search was conducted on July 25, 2024, and subsequently updated on June 20, 2025, to ensure the inclusion of the most recent literature. The detailed search strings used for each database are reported in Supplementary Table S1.

2.2. Eligibility Criteria

The selection criteria were defined a priori. We included only systematic and scoping reviews that address the impact of climate change on health and safety among the occupational population. Only articles published in English were considered, with no restrictions applied to the publication year. Eligible reviews had to investigate one or more of the following domains: (i) health and safety effects attributable to climate-related occupational exposures and (ii) effectiveness of interventions aimed at reducing health and safety impacts.

We excluded narrative reviews, those focused solely on heat or cold exposure in occupational settings not explicitly linked to climate variability, those considering professional athletes only, and those considering economic consequences without reporting any health or safety outcomes. We also excluded

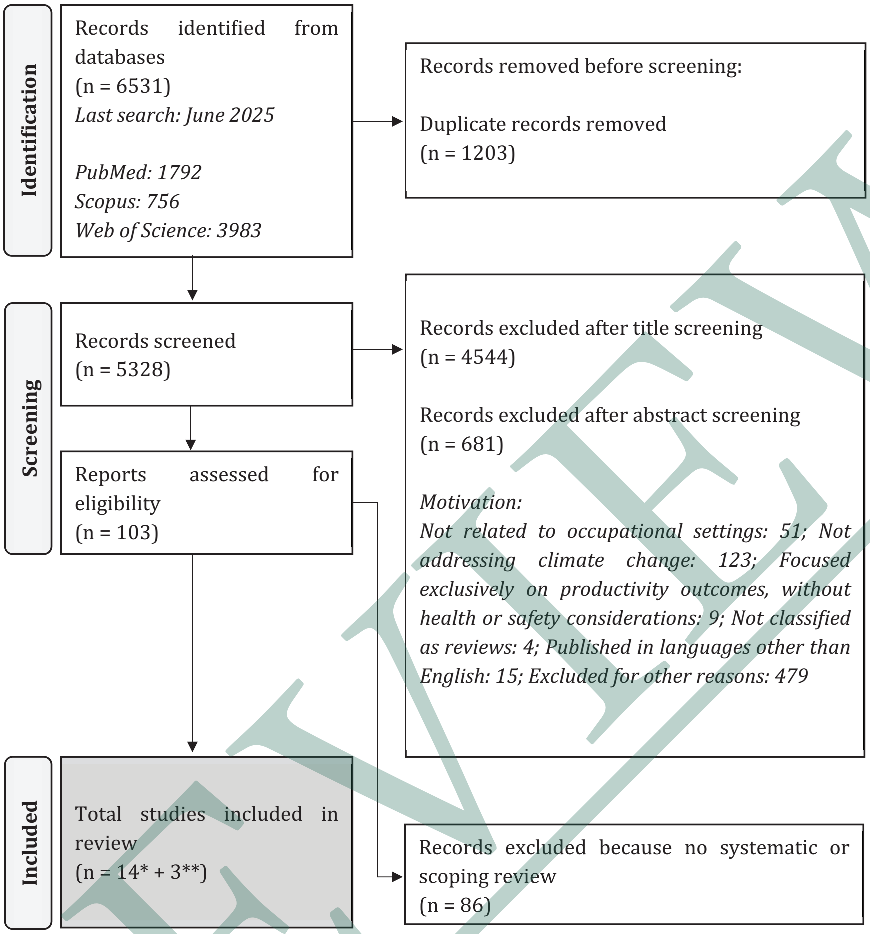

Figure 1. PRISMA Flowchart.

**14 reviews on health and safety effects; ** 3 reviews on intervention to prevent heat-related health and safety effects.*

reviews exclusively investigating secondary effects such as radiation or pesticide exposure, or focused on very specific populations (i.e., a single country).

Finally, in line with the study aims, we focused exclusively on reviews addressing the health and safety effects of heat exposure; reviews investigating the potential implications of milder winters or other seasonal temperature changes were excluded, as these phenomena fall outside the defined scope of this umbrella review.

2.3. Selection Process

After duplicates removal, the screening process was conducted using Rayyan, a web-based tool for systematic review screening. Two independent reviewers screened titles and abstracts to identify potentially relevant articles. Disagreements were resolved through discussion with a third author. Articles deemed relevant at this stage were retrieved in full text and independently assessed by at least two of the authors. All included reviews were read in full and screened in duplicate. Discrepancies in inclusion or data extraction were resolved by consensus among the reviewers.

2.4. Data Extraction

Data extraction was performed using a structured Microsoft Excel sheet. The following information was collected for each included review (and reported in Table 1 and Tables S2-S4): author(s) and year of publication; country of the corresponding study or

Table 1. Characteristics of reviews on health effects of climate change by review type. * The review also carried out a meta-analysis. WBGT: Wet Bulb Globe Temperature; TWL: Thermal Work Limit.

Systematic Reviews							
Author, Year, Country	**Design, Aims**	**Databases Search, Search Period**	**Eligibility Criteria**	**Occupational Settings Considered**	**Exposure**	**Primary Outcome**	**N Studies**
Binazzi et al., 2018 Global	Summarizing the epidemiological evidence of the associations between rising heat due to climate change and occupational injury risk. Identify occupational sectors at increased risk of heat stress.	PubMed; EMBASE January 2000 – October 2018	• English language. • Epidemiologic studies published from 01 January 2000 until 03 October 2018. • Papers with a case-crossover or time-series design. • Studies involving humans (men or/and women). • Studies focused on heat-related injuries among workers, with special attention to the construction and agricultural sectors. • Studies with available meteorological data. • Studies providing effect estimates with the corresponding measures of variability.	Workers (in general) with special attention to the construction and agricultural sectors	Daily maximum temperatures Daily minimum temperatures WBGT. Other heat stress indices. Heat wave classifications.	Occupational injury risk	8 (3 case-crossover studies; 5 time-series studies)
Fatima et al., 2021 Global	To assess how a 1°C increase in temperature, including hot temperatures and heatwaves, affects the incidence of occupational injuries by comparing heatwave periods to non-heatwave periods. To analyze how the risk of occupational injuries associated with extreme heat varies across different climate zones around the world. To identify which groups of workers, based on their characteristics, the nature of their work, and workplace features, are at higher risk of heat-related occupational injuries.	PubMed; Embase; Scopus Up to July 2020	• Population: all studies assessing the risk of occupational injuries among non-military workers associated with extreme heat, published from database inception until July 20, 2020. • Exposure: all epidemiological studies where extreme heat was quantified in terms of hot temperature and heatwaves. • Comparator: study design did not use any comparators, however the risk is estimated with reference to lower level of exposure. • Outcome: occupational injuries. • Study Design: observational studies including ecological studies, prospective and retrospective cohort studies and case-control studies were considered for inclusion.	Non-military workers	Maximum temperature (Tmax) Minimum temperature (Tmin) Mean temperature (Tmean) WBGT Apparent temperature (AT) Humidex Heatwaves classifications	Occupational injury risk	24 (22 for meta-analysis)

Florez-Acevedo et al., 2025 Global	Review primary research studies on the relationship between occupational heat exposure and mental health outcomes. Synthesize the literature to identify gaps and opportunities for future research with a proposed framework that takes into account social determinants of health (SDOH), in addition to modifiable workplace physical and psychosocial factors, using an equity lens.	PubMed; Medline; Web of Science; Scopus; PsycINFO 1997 – 2024	• Primary research studies published in peer-review journals. • Qualitative or quantitative observational, quasi-experimental, or experimental study designs. • Assess mental health and heat exposure in an occupational setting.	Workers (in general)	WBGT	Mental health outcomes	10
Flouris et al., 2018 Global	Estimated the effects of occupational heat strain on workers' health and productivity outcomes.	PubMed; EMBASE Manual searches for published studies in international trial registers and websites of international agencies Up to February 2018	• Articles consisted of original quantitative research. • Articles published in a peer-reviewed journal or scholarly report.	Workers (in general)	WBGT Air temperature	Health impacts Productivity	111 studies included in qualitative analysis 64 studies included in meta-analyses by outcome
Habibi et al., 2021 Global	Collect information on the impacts of climate change on occupational heat stress in outdoor workers. Identify factors that increase worker susceptibility to heat stress. Report strategies to manage and reduce climate change impacts on occupational heat stress. Provide guidance for employers, stakeholders, occupational hygienists, healthcare professionals. Support the development of mitigation policies and practical solutions for prevention and control.	PubMed; Scopus; Web of Science To find more relevant studies the reference list of included studies was investigated 2000 – October 2019	• Peer reviewed articles. • Paper conducted on men and women workers. • Paper about climate change, heat waves, UV radiation, and heat stress exposure. • Paper on the impacts of climate change on heat strain. • Paper consisted of comparison among countries or workplaces. • Paper conducted on working populations when facing climate change and solar radiation.	Outdoor workers	-	Occupational heat strain	25

Table 1 (*Continued*)

Systematic Reviews							
Author, Year, Country	**Design, Aims**	**Databases Search, Search Period**	**Eligibility Criteria**	**Occupational Settings Considered**	**Exposure**	**Primary Outcome**	**N Studies**
Habibi et al., 2024 Global	Assess the effect of climate change on occupational heat strain among women workers.	PubMed; Scopus; Web of Science Reference list of included studies 2000 - October 2019	• Peer-reviewed articles. • Papers conducted on women workers. • Papers about climate change. • Paper that outlined the impacts of climate change on heat strain. • Article consisted of comparisons between countries or workplaces.	Women	-	Health impacts Heat strain Risk factors that may increase susceptibility to climate-related occupational hazards Controlling factors of climate change and occupational heat strain	13
Levi et al., 2018 Global	Part of HEAT-SHIELD project funded by the European Union dedicated to address the negative impact of increased workplace heat stress on the health and productivity of five strategic European industries: manufacturing, construction, transportation, tourism and agriculture. Summarize the epidemiological evidence of the effects of climate change, with a special focus on high temperatures and heat waves, on workers' health and productivity, in order to better inform health policies in the EU and beyond. Identify size and severity of the existing relationship between heat-related to climate change and workers' health and productivity.	PubMed; EMBASE; Scopus January 2000 – June 2017	• Studies published from 01/01/2000 until 16/06/2017. • Studies focused on heat-related illness and injuries among workers or on workers' productivity with special attention to the construction and agricultural sectors.	Workers (in general)	Daily Maximum Temperature WBGT Average annual summer temperature (Tmax) Apparent Temperature Hourly humidity Hourly maximum temperature Mean maximum temperatures Maximum heat index	Health impacts Occupational injury risk Productivity	104 (heat-related illness, cardiovascular, respiratory and kidney diseases) 15 (traumatic injuries and acute death) 37 (vector-borne diseases or vectors distribution) 28 (effects of climate change on work productivity)

Tang et al., 2025 Global	This systematic review aims to examine how climate change and its related stressors may affect the mental health of workers in industries vulnerable to climate change. The review also seeks to evaluate coping strategies used by affected workers, as well as potential interventions to mitigate and prevent these mental health effects.	PubMed; Embase; PsycINFO; Web of Science Up to 1st June 2024	• Workers in industries vulnerable to climate change. • Study assesses climate change or climate related stressors against mental health related outcomes, associated risk factors or assesses interventions or coping strategies for mental wellbeing against climate related stressors. • The study is a peer reviewed study. • Article written in English.	Workers in: Agriculture Manufacturing Construction Industry Energy-Generating Resources Transportation	-	Mental disorders Mental health	42
Varghese et al., 2018 Global	Summarise the evidence on the relationship between heat exposure and injuries. Describe aetiological mechanisms. Provide policy suggestions and further research directions.	PubMed; Embase; Scopus; CINAHL; Science Direct; Web of Science Grey literature (unpublished studies were searched in internet search engines and web-based searches) Until January 2017	• Original research articles in English published until 31st of January 2017. • Studies which investigated the association between heat exposures and work-related injuries/accidents.	Workers (in general)	Maximum temperature Minimum temperature Apparent temperature Heat Index Humidex WBGT	Occupational injury risk	26 (4 of which unpublished)

Scoping Review, Mapping Review

Author, Year, Country	Design, Aims	Databases Search, Search Period	Eligibility Criteria	Sample Characteristics	Exposure	Primary outcome	N studies
Acharya et al., 2018 USA/global	*Scoping review* Review the existing epidemiological research about occupational heat stress in the construction industry, both in the U.S. and internationally. Investigate the risk factors for heat-related illnesses in construction, as well as populations most vulnerable to experiencing heat-related illness. Review preventive measures and existing worker protection policies enacted by local and national governments.	PubMed; Web of Science References of relevant peer-reviewed literature Web-based searches (including Google Scholar and documents published by occupational-health organizations) Up to March 2017	• Papers that addressed or analyzed heat stress exposure, risk of heat-related ilness, or interventions for heat-related illness among construction workers.	Construction workers	Temperature Temperature max WBGT Solar radiant heat TWL Humidity	Health impacts Occupational injury risk	16

Table 1 (*Continued*)

Author, Year, Country	Design, Aims	Databases Search, Search Period	Eligibility Criteria	Sample Characteristics	Exposure	Primary outcome	N studies
Scoping Review, Mapping Review							
Amoadu et al., 2023 Global	*Scoping review* Find risk factors that increase workers' susceptibility to heat stress. Unveil the impact of climate change on workers' health and productivity. Find the measures used in controlling heat stress among workers.	PubMed; Central; Scopus; Web of Science In addition to the main databases, searches were conducted in other databases such as JSTOR, Google Scholar, Google, and Hinari, and institutional repositories also were explored to identify relevant articles. Furthermore, experts (review expert and a librarian) were consulted and reference lists were checked to find relevant additional studies for this review January 2005 - September 2022	• Outcome: studies assessing: (1) workers' health outcomes related to heat stress; (2) workers' productivity outcomes related to heat stress; (3) risk factors associated with workers' health and productivity outcomes; and (4) adaptation measures. • Population: all occupational groups of all age groups and sex. • Study design: qualitative studies, mixed methods studies and quantitative studies, including studies that calculated the effects with simulations or models. • Language filter: studies written in the English language only. • Time filter: studies that were published online in 2005 and later.	Workers (in general)	-	Health impacts Productivity	157

El Khayat et al., 2022 Global	*Scoping review* Summarize the available evidence on the effects of climate change on farmworkers' health, focusing on heat-related illnesses. Identify the risk factors for heat-related illnesses among farmworkers. Review the preventive measures used to minimize heat stress exposure among farmworkers. Inform policymakers so they can develop more effective policies and programs to protect vulnerable farming communities from the effects of climate change.	Medline; Embase; Scopus; CINAHL; Web of Science Gray literature The websites of international agencies (ILO,WHO, and UNDP) and agricultural databases (AGRIS FAO and AgEcon) were manually searched to identify relevant articles Up to December 2021	• Outcome of interest: Studies assessing: (1) Health outcomes related to heat exposure; (2) Risk and protective factors associated with health outcomes; and (3) Preventive measures. • Population of interest: Farmworkers of all age groups. • Context of interest: Studies assessing the impact of occupational heat exposure on workers' health and safety. • Study design: Primary studies including quantitative and qualitative methods.	Farmworkers	Temperature Relative humidity WBGT Heat index Heat stress index Humidex Universal thermal climate Index	Health impacts Occupational injury risk	92
Ferrari et al., 2023 Global	*Mapping review* Explores how the impacts of climate change are related to workers' health and safety. Investigates the methodologies used to assess the effects of climate change on workers' health and safety. Identifies the main negative effects of climate change on workers.	Emerald; IEEE Xplore; Science Direct; Scielo; Scopus; SpringerLink; Web of Science Snowballing process 2010 – 2020	• Articles that explore the adverse effects on workers' health caused directly by some effect of climate change. • Articles that address a loss of productivity or economic impact due to impacts on workers' health caused by some effect of climate change. • Articles that discuss scenarios or projections of the effects of climate change on workers' health.	Workers (in general)	Temperature variation Periods of extreme heat Heatwaves Cold WBGT	Health impacts Occupational injury risk	170
Guihenneuc et al., 2023 Global	*Mapping review* Highlight the components of health care facilities affected by climate change.	1st aim: PubMed 2nd aim: PubMed; Scopus; Science Direct; Cairn; GreenFile (and papers identified from the reference lists of included papers) 1st aim: 1979-2021 2nd aim: 1979-2019	• 2nd aim (mapping review): • Studies published between January 01, 1979 and December 31, 2019. • Studies that identified the presence of at least one effect on one of the health care facilities components by one of the climate drivers.	Manpower (staff working in health care facilities and all the patients who could be cared for)	-	Health facility vulnerability	46 (mapping review)

Table 2. Methodological limitations identified in current systematic reviews and suggested future research developments.

Identified Limitation	Suggested Future Development
Heterogeneity of study designs	Develop harmonized protocols and guidelines for occupational climate–health and safety studies.
Variability in exposure indicators	Standardize exposure metrics and promote validation of measurement tools.
Inconsistent definitions of health outcomes	Create shared taxonomies and outcome definitions for comparability across studies.
Use of indirect exposure data	Increase use of personal/wearable sensors and high-resolution environmental data.
Paucity of longitudinal studies	Conduct long-term cohort and follow-up studies to assess chronic and cumulative effects.
Limited attention to vulnerable groups	Design targeted studies addressing underrepresented populations and contexts.
Inadequate control of confounders	Integrate individual-level data and improve multivariable adjustment models.
Scarce attention to less frequently studied outcomes	Expand research focuses to emerging outcomes with interdisciplinary approaches.
Fragmented disciplinary approaches	Foster collaboration across climatology, epidemiology, occupational hygiene, and social sciences.
Limited translation into practice	Develop evidence-based prevention interventions and policy guidelines for workplaces.

region of interest; review type and study design; databases searched and search period; eligibility criteria applied; type of occupational population investigated; exposure metrics and climate-related variables assessed; primary outcomes and number of studies included; aims of the review; types of workers and outcomes actually examined; preventive interventions and mitigation strategies or best practices reported; limitations of the included studies; research gaps and future directions; authors' main conclusions.

To facilitate a structured narrative synthesis, the extracted information was then organized into thematic tables: Table 1: methodological characteristics of the included reviews; Table S2: results in terms of study populations and main identified health and safety outcomes; Table S3: identified limitations, future perspectives, and conclusions; Table S4: suggested policy recommendations and adaptation measures; and Table 2:methodological limitations identified in current systematic reviews and suggested future research developments. This approach allowed grouping evidence by outcome, population, and intervention level, thus supporting a clearer comparative analysis across reviews. Since only a limited number of reviews of intervention studies were retrieved, we reported their characteristics and results in separate Tables (Table S5-6).

2.5. Risk of Bias Assessment

The ROBIS tool [9, 10] was used to assess the methodological quality of systematic reviews considered in this study. This evaluation was independently performed by two reviewers, and any discrepancies were resolved by consensus. The ROBIS tool enabled assessment of potential bias in study selection, data collection, synthesis methods, and interpretation of findings.

The risk-of-bias assessment was applied only to systematic reviews and was therefore not performed for other types of studies (i.e., scoping and mapping reviews).

3. Results

As reported in Figure 2 and in Figure S1, ROBIS judgments [9, 10] indicated that the overall risk of bias was predominantly high, with 90% of reviews classified as high risk and only 10% as low risk. Domain-specific ratings were more heterogeneous: all reviews were judged at low risk in D1

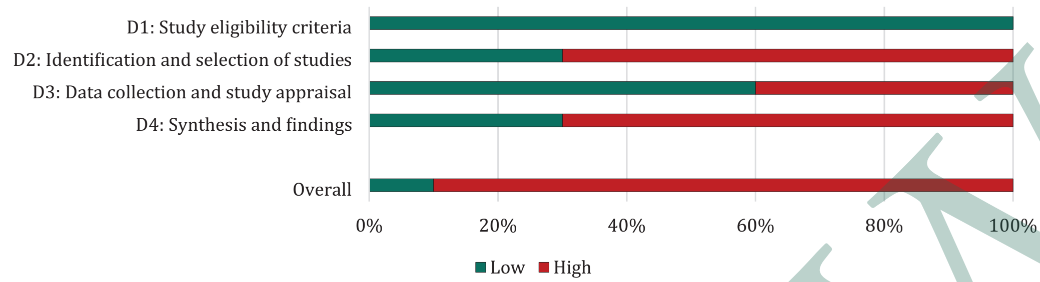

Figure 2. Summary of the evaluation of risk of bias in included reviews using ROBIS tool.

(study eligibility criteria), while D3 (data collection and study appraisal) was rated low risk in 60% of reviews and high risk in 40%. In contrast, most reviews were judged to be at high risk in D2 (identification and selection of studies) and D4 (synthesis and findings), with about 70% of reviews falling into the high-risk category in each domain.

It is important to note that some reviews received an overall high-risk-of-bias rating solely because a single critical domain was assessed as high risk (Figure S1), rather than due to consistent methodological weaknesses across domains. Others, instead, showed multiple unclear or high ratings across different areas, underscoring that the overall classification often reflected isolated issues rather than a pervasive bias. This distinction is relevant for interpreting the robustness of the evidence, as reviews affected by isolated critical issues may still provide partially reliable findings, whereas those with multiple high-risk domains require substantially greater caution in interpretation.

3.1. Reviews on Health and Safety Effects

3.1.1. Design Characteristics of Identified Reviews

We identified 14 reviews of scientific literature examining the impacts of climate change on the health and safety of the working population (Table 1).

Methodologically, approaches varied: (i) nine systematic reviews (three incorporating a meta-analysis); (ii) three scoping reviews, and (iii) two mapping reviews. The number of primary studies per review ranged from 8 to 170. The time frame covered by the included primary studies spanned 1979-2024, with a marked increase in publications over the past decade. In particular, the overall objectives of the included reviews varied, ranging from: (i) summarizing epidemiological data on the relationship between heat and occupational injuries; (ii) evaluating the effects of heat and heatwaves on workers' health and safety; and (iii) investigating emerging issues, like the psychological impact of high temperatures. The populations studied were the general workforce in some instances, while other reviews focused on specific outdoor sectors (e.g., agriculture, construction). Some reviews explicitly excluded military personnel, focusing instead on civilian workers and standard occupational settings. The exposure metrics varied, including both composite indices and direct readings. Commonly used indicators included Wet Bulb Globe Temperature (WBGT) and peak temperatures. The outcomes analyzed referred to: (i) occupational injury risk, often identified as the key consequence of heat exposure, (ii) heat stress, and (iii) general health impacts, such as selected acute and chronic conditions.

3.1.2. Main Results of Identified Reviews

Occupational Groups Considered in the Reviews

The included reviews synthesized evidence across a wide variety of occupational categories (Table 1). The groups most frequently considered were construction (8 reviews) and agricultural workers (7 reviews), as well as miners (5 reviews) and, more generally, outdoor workers (4 reviews). In addition, less frequently covered categories included industrial and manufacturing workers and, in some cases, more specific groups (glass, aluminum, aquaculture, aviation, logistics), as well as public-service roles (law enforcement, firefighters, paramedics, healthcare workers). Finally, some reviews reported evidence regarding indoor workers (2 reviews).

Health and Safety Impacts of Climate Change on Workers

The reviews documented a wide range of effects associated with workers' exposure to extreme climatic conditions (Table S2). Among these, occupational injuries were one of the most frequently reported outcomes, along with heat-related illnesses in general. In particular, recurrent clinical outcomes include kidney disease (10 reviews), respiratory illnesses (6 reviews), dehydration (6 reviews), and mortality (6 reviews). Psychological symptoms and mental health disorders also emerge as significant (6 and 5 reviews, respectively). Other reported outcomes include cardiovascular disease, allergies, and vector-borne diseases (4 reviews), while more specific but less frequent effects include headache, hypertension, eye problems, traumatic injuries (wounds, falls, fractures), and skin, gastrointestinal, and reproductive disorders. Individual cases of serious conditions such as heart attack, cancer, suicide, and infectious diseases (e.g., typhoid) have also been reported. Although it is outside the scope of this work, it is important to underline how some reviews highlighted indirect consequences, such as reduced labor productivity, health-system challenges in managing demand peaks, and increased vulnerability among high-risk worker groups.

Methodological Challenges and Research Gaps

The included systematic reviews highlighted several common methodological limitations (Table S3 and Table 2). In particular, a marked heterogeneity among primary studies was observed in terms of (i) study design, (ii) exposure indicators used, and (iii) definitions of health and safety outcomes. Further critical issues include the frequent use of indirect exposure data (e.g., weather station metrics, self-reported questionnaires). Furthermore, assessments rarely stratified vulnerable subpopulations, such as women, migrants, indoor workers, and residents of low- and middle-income countries, as well as the lack of systematic consideration of individual confounding factors (chronic diseases, alcohol consumption, medications). Finally, many reviews highlighted the limited attention to less frequently studied outcomes, such as mental health, renal disease, and productivity loss.

The main future research directions suggested were: (i) standardize exposure metrics and assessment tools; (ii) conduct longitudinal and longer-term studies; (iii) investigate emerging outcomes, such as mental health, kidney disease, and work performance; (iv) adopt interdisciplinary approaches integrating climatology, epidemiology, medicine, and occupational hygiene; (v) develop targeted studies and interventions in less-explored populations and contexts; and (vi) promote practical solutions and evidence-based prevention interventions.

Overall, the reviews' conclusions converge in recognizing occupational heat exposure as a concrete and growing risk to workers' health and safety. The evidence is particularly strong for injuries and heat strain, especially in outdoor sectors such as agriculture and construction, while growing concerns are also emerging regarding the effects on mental health and chronic diseases. Finally, the reviews emphasize the urgent need to implement integrated mitigation interventions and preventive strategies at the individual, organizational, and policy levels to reduce the current and future impacts of climate change on the workplace.

Suggested Preventive Interventions

The reviews suggested a wide range of strategies aimed at mitigating the impacts of climate change and, in particular, occupational heat exposure, on workers' health and safety (Table S4). These measures can be generally grouped into different levels of intervention:

1. Training and communication: disseminating knowledge about heat risks, early symptoms, and preventive practices among workers and employers, with active involvement of occupational physicians and other health professionals.

2. Managerial controls: reorganizing working hours (e.g., shift work, job rotation, mandatory breaks during the hottest hours, suspension of activities above certain temperature thresholds), developing company guidelines, and training programs for workers.
3. Engineering and environmental controls: implementing structural solutions to reduce heat exposure, such as shading, ventilation, and cooling systems, thermal insulation, use of fans, provision of air-conditioned rest areas, and assured access to fresh drinking water.
4. Individual measures: adequate hydration, use of suitable clothing and personal protective equipment – PPE (e.g., hats, technical fabrics), self-regulation of work pace, and awareness of early signs of heat stress.
5. Standards and surveillance systems: establishing local exposure thresholds, adapting climate conditions and specific tasks, implementing health surveillance programs for the early identification of heat-stress effects, and integrating weather-warning systems for safe activity planning.
6. Regulatory framework: implementing national and international guidelines, deploying weather-warning systems for employers and workers, and enforcing legislation that includes mandatory training and medical screening.
7. Targeted approaches for vulnerable groups: welfare measures and economic support, especially in the most fragile contexts, with emphasis on migrant workers and women.

3.2. Reviews On Interventions to Prevent Heat-Related Health And Safety Effects

Three reviews that specifically addressed interventions to prevent heat-related health and safety effects in occupational settings were identified (Table S5 and Table S6). These included one systematic review with meta-analysis focusing on construction [11], one scoping review with a broader cross-sectoral perspective [12], and one systematic review on agricultural workers [13]. The number of studies included in these reviews range from 9 to 69. The populations investigated covered construction workers, agricultural workers, and more general categories of workers.

The reviews consistently highlighted the importance of multifaceted approaches to heat stress prevention. In construction, Esfahani and collaborators [11] clustered risk factors into demographic, individual, environmental, social, regulatory, occupational, organizational, and PPE-related dimensions. Their meta-analysis confirmed the effectiveness of cooling vests and specialized uniforms in significantly reducing core body temperature and heat strain. Khorshid and Song [12] emphasized that the evidence is strongest in the sports sector but transferable to occupational contexts, demonstrating clear benefits of both internal and external interventions. Moreover, they found that combined strategies were reported to produce greater benefits than single interventions in the reviewed studies; however, no pooled effect estimates were available across reviews, and the generalisability of these findings to different occupational sectors, climatic contexts, and resource settings may remain uncertain. Finally, Roca and collaborators [13], focusing on agricultural workers, reported that educational interventions improve knowledge and behavior, while hydration and body-cooling devices reduce heat strain.

It should be noted that the evidence base for occupational heat interventions, as captured by this umbrella review, remains limited. The three identified reviews were heterogeneous in design, population, and outcome measures and did not permit quantitative synthesis of effect sizes at the umbrella level. Findings should therefore be interpreted as indicative of promising directions rather than as definitive evidence of effectiveness.

4. Discussion

This work highlights that the relationship between climate change and occupational health and safety is widely recognized as a global concern [4, 14-15]. The international perspective adopted by all included reviews emphasizes the cross-cutting nature of this phenomenon, whose effects may vary across contexts, being more or less pronounced depending on specific vulnerabilities and adaptive capacities. Moreover, the marked increase in publications over the past decade reflects growing awareness within the scientific community of the need to better understand the consequences of climate change on workers' health and safety.

From a methodological perspective (Table 1), the heterogeneity of study designs, exposure metrics, and outcome measures substantially limits the possibility of direct comparisons between results [1-3, 16-17]. Only a few reviews included a meta-analysis [2, 3, 14], while several relied on scoping or mapping approaches, reflecting the still exploratory nature of this research field. It is important to emphasize that the present umbrella review is based on a qualitative synthesis of heterogeneous evidence. As such, the conclusions drawn are not supported by pooled quantitative estimates and should be interpreted with caution. While the identified patterns are consistent across reviews, the lack of quantitative aggregation may limit the strength of inference and the level of evidence available for decision-making. The limited number of meta-analyses can be attributed to multiple factors: (i) substantial heterogeneity in exposure definitions and measurement: while some studies adopt WBGT, others rely on maximum daily temperature, heat index, or alternative composite metrics-resulting in non-comparable datasets; (ii) inconsistent outcome categorization (e.g., variable definitions of heat-related illnesses, injuries, productivity loss) reducing the feasibility of quantitative pooling; (iii) a primary evidence base dominated by observational designs, which often lacks the standardized, high-quality data required for robust synthesis; and (iv) wide variability in geographical settings and occupational groups, reflecting differences in climate, socio-economic context, and working conditions, further complicating aggregation. Taken together, these elements may explain why most authors opted for scoping or narrative syntheses rather than attempting a meta-analysis.

A more detailed interpretation of the risk of bias assessment is warranted. Although the majority of reviews were classified as having a high overall risk of bias (9 out of 10), this classification did not reflect uniformly poor methodological quality. In several cases, the high-risk judgment was driven by limitations in a single domain, most commonly related to study identification and selection. Conversely, other reviews presented weaknesses across multiple domains, including data collection and synthesis, indicating more pervasive methodological limitations. This distinction is important, as it directly influences the confidence that can be placed in the synthesized evidence. Reviews affected by isolated issues may still provide partially reliable insights, whereas those with more widespread methodological concerns require substantially greater caution in interpretation.

Analysis of the occupational categories considered in the reviews highlights how the impacts of climate change are not uniform and vary significantly depending on the sector, type of job [3, 6] and working environment (Table S2). The increased focus on construction and agricultural workers reflects their distinct vulnerability, driven by the combination of physically demanding tasks and prolonged exposure to the outdoor environment [2, 4, 16, 18]. In both sectors, several factors amplify climate-related risks: high levels of physical exertion [1, 4, 18], direct and prolonged exposure to extreme temperatures [4, 15-16, 18], and, in many cases, limited access to adequate protective measures (e.g., shading, ventilation/cooling systems, scheduled hydration breaks) [1, 16, 18-19]. Moreover, in agriculture, workers are often engaged in seasonal tasks performed in rural or remote settings [16], where access to healthcare and emergency support may be restricted, further increasing vulnerability [1, 16]. By contrast, construction workers typically operate in urban environments where the urban heat island effect, exposure to asphalt and concrete surfaces, and co-exposures to pollutants can compound the risks associated with heat [4, 18]. In both cases, precarious forms of employment, informal work arrangements, and a lack of systematic training on preventive measures can exacerbate the impacts, highlighting that climate-related vulnerability is shaped not only by environmental conditions but also by social, economic, and organizational dimensions [1, 16, 18-19]. Similarly, miners and, more generally, outdoor workers emerge as particularly at-risk groups, which is consistent with evidence that work performed in extreme environmental conditions amplifies heat-related hazards. More in detail, urban workers (e.g., construction, traffic police, logistics) are potentially subject not only to high ambient temperatures but also to the additional burden of urban heat-island effects [19], limited ventilation [2, 19], reduced access to shaded or green areas [18], and co-exposures such as air pollution [15, 20], noise, and traffic-related stressors. Conversely, non-urban workers (e.g., agricultural and forestry workers) may face fewer artificial modifiers, but they are often exposed to longer, more direct, and less controllable climatic extremes, including prolonged sun exposure [1,15-16], dust [4, 18], pesticides [1, 15] and vector-borne hazards [4, 15]. Furthermore, the remoteness of many rural workplaces can hinder timely access to medical care, adequate hydration, and rest facilities, amplifying the risk of severe outcomes [1, 16]. However, the fact that indoor workers were also considered in some reviews indicates that risks are not confined exclusively to outdoor activities [2]. Even poorly ventilated or inadequately air-conditioned indoor environments can pose challenges, suggesting that indoor workers may have been neglected, and that vulnerability must be assessed by taking into account specific working conditions and available adaptation measures [1, 6, 15].

The analysis of vulnerability to heat stress should extend across both outdoor and indoor sectors, including agriculture, mining, public utilities, and manufacturing [4, 6, 17-18]. Gender-related vulnerability to heat stress emerges as a multidimensional phenomenon, shaped by physiological differences, socio-organizational constraints, and cultural factors. Although men often account for a higher absolute number of heat-related injuries due to their concentration in high-intensity sectors such as construction and mining, women face distinct and often underrecognized risks. Gender-related differences represent a key dimension of differential vulnerability to occupational heat exposure. Women exhibit distinct physiological responses to heat, including higher sweating thresholds and lower sweat rates, which may reduce the efficiency of evaporative cooling and lead to greater internal heat storage [20]. Differences in body composition and hormonal fluctuations across the menstrual cycle and menopause may further influence thermoregulation and heat tolerance [20]. Pregnant workers represent a particularly vulnerable subgroup, as pregnancy-related physiological changes may further impair thermoregulation and increase susceptibility to adverse outcomes, including fetal distress, preterm birth, and other complications [1, 20]. In addition to physiological factors, socio-organizational conditions play a critical role. In several occupational settings, particularly in agriculture and construction, women may intentionally limit fluid intake to avoid using distant, inadequate, or unsafe sanitation facilities, increasing the risk of dehydration and renal disorders, including acute and chronic kidney disease [1, 16]. This phenomenon may be exacerbated under piece-rate payment systems, which incentivize reduced breaks and delayed hydration. Structural and cultural factors may also amplify heat-related risks. In some contexts, clothing requirements or cultural norms may limit heat dissipation, while the use of personal protective equipment not designed for female body characteristics may increase physical strain and thermal discomfort [1, 20]. Finally, although women are often underrepresented in the most physically demanding occupations, available evidence suggests that they remain exposed to significant heat-related risks, including injuries associated with fatigue, dizziness, and impaired concentration during heat events [3, 6, 18]. Migrant workers and ethnic minorities represent highly vulnerable populations with respect to occupational heat exposure, due to the interplay of socio-economic disadvantages, legal constraints, and demanding working conditions. These workers often face limited decision-making autonomy and reduced control over workplace safety, as their legal or employment status may discourage them from reporting hazardous conditions or requesting adequate breaks and hydration [1, 16-17]. Language and cultural barriers may further hinder the understanding of safety training and early warning signs of heat-related illness, while cultural norms may influence risk perception and discourage the expression of discomfort [3, 16-18]. Migrant workers are frequently overrepresented in high-intensity sectors such as agriculture and construction, where physical exertion and outdoor exposure are particularly high [5, 14, 18]. Newly arrived workers may also lack adequate acclimatization to local climatic conditions, increasing the risk of heat-related illness and injury, particularly during the initial period of employment [3, 5, 18]. In addition, vulnerability may extend beyond the workplace, as many migrant workers live in substandard housing or urban heat island areas without adequate cooling, limiting physiological recovery during rest periods [18]. These factors may contribute to marked disparities in health outcomes. Higher rates of heat-related mortality have been reported among specific ethnic groups, such as Hispanic, Black, and Native American workers in the United States [18]. Migrant workers are also disproportionately affected by chronic kidney disease of unknown origin, associated with repeated dehydration and heat stress in sectors such as agriculture [5, 16], as well as by infectious diseases linked to climate variability [5]. Limited access to healthcare services and occupational protections further exacerbates these risks. Migrant workers may face barriers to healthcare access and compensation systems [1, 18], and in some contexts may be excluded from basic labor protections, including access to shade, drinking water, and regulated work-rest cycles [16]. Addressing these vulnerabilities requires culturally and linguistically appropriate interventions, as well as occupational health surveillance systems that explicitly consider social determinants of health [1, 14, 16, 20]. Age-related vulnerability to heat stress emerges as a pattern of "dual risk", disproportionately affecting both younger and older workers through different mechanisms [1, 3, 18, 20]. Younger workers (typically under 25-35 years) often report higher rates of heat-related injuries and acute conditions [3, 6, 18]. Their vulnerability is largely driven by behavioral and organizational factors. They are more likely to be assigned physically demanding tasks [2-3], may have limited training and experience in recognizing early signs of heat stress [2, 15, 20], and often exhibit lower risk perception and adherence to preventive measures, sometimes influenced by peer pressure or workplace norms [2-3, 15]. In addition, newly employed workers are frequently not adequately acclimatized to heat, which substantially increases the risk of injury during the initial period of exposure [6]. In contrast, older workers (typically over 45-65 years) tend to experience higher mortality rates during heat events [14, 18], reflecting predominantly physiological vulnerability. Age-related declines in thermoregulatory capacity, including reduced sweating efficiency and skin blood flow, impair heat dissipation [3, 20]. This vulnerability is further exacerbated by the higher prevalence of chronic conditions such as diabetes, hypertension, and cardiovascular or respiratory diseases [3, 14-15, 18]. Among older women, menopause-related hormonal changes may further alter thermoregulation and increase heat storage [20]. Moreover, older workers appear more likely to experience severe outcomes following heat-related incidents, as suggested by higher insurance costs associated with injuries during heat waves, indicating reduced recovery capacity [6]. Finally, piece-rate payment systems and precarious socio-economic conditions act as strong drivers of vulnerability. Piece-rate work may incentivize workers to ignore physiological warning signs and skip hydration breaks in order to maximize earnings, while low income is often associated with substandard housing in urban heat islands, limiting adequate recovery during rest periods [1, 5, 16, 18].

Regarding health outcomes, the reviews confirm the complex and multifaceted impacts of climate change on workers' health and safety (Table S2). Occupational injuries and heat-related illnesses are the most frequently reported outcomes, demonstrating how rising temperatures simultaneously affect both safety and health [3-4, 17]. The broad spectrum of reported clinical outcomes (from renal and respiratory diseases to dehydration and mortality) underscores the potential severity of heat exposure. At the same time, the emergence of psychological symptoms and mental health disorders highlights a broadening of the investigative focus, progressively moving beyond immediate physical effects to include less tangible yet equally relevant dimensions of workers' well-being. For several outcomes, cardiovascular and allergic diseases, heart attack, cancer, or suicide, climate change may function as a multiplier of pre-existing risks and as a cross-cutting vulnerability factor for various organ systems. Finally, although beyond the scope of this review, several studies pointed to the combined role of co-exposures such as ultraviolet radiation, air pollution, high physical workloads, pesticide use, or chemical agents, which may exacerbate the impacts of climate stressors. In this umbrella review we focused exclusively on heat-related effects in occupational populations and did not explicitly consider that climate change may also lead to milder winters and potentially reduce cold-related health impacts. Evidence from general-population studies, however, suggests that the balance between heat- and cold-related risks may vary across countries and climatic contexts [21-24].

The systematic reviews included in this work highlighted a set of methodological limitations that deserve consideration (Table S3 and Table 2). A first observation is the preponderance of narrative and scoping reviews relative to full systematic reviews, and the paucity of meta-analyses. This aspect may reflect structural barriers that hinder quantitative syntheses, hampering data harmonization and comparability: (i) marked heterogeneity in study designs [6, 15, 17, 19-20]; (ii) variety of exposure indicators (WBGT, daily maximum temperature, heat index, composite indices) [2-3, 5, 16, 18]; and (iii) inconsistent definitions of health outcomes (from general indicators to more specific diagnoses) making it difficult to align evidence across studies and to develop a coherent picture of health risks [1-3, 6, 19]. It is likely that, at least in the short term, this fragmentation will continue to limit comparability, calling for a greater effort toward methodological standardization. Another critical issue concerns the quality of primary exposure data, which are often derived from indirect measures such as coarse meteorological indicators or self-reported questionnaires. These approaches can lead to misclassification and undermine the robustness of conclusions [2, 3, 6]. An additional limitation concerns the potential overlap of primary studies across the included reviews. As multiple reviews may rely on partially shared datasets, some findings could be overrepresented in the synthesis. This inherent characteristic of umbrella reviews should be considered when interpreting the consistency and weight of the available evidence. In addition, the limited number of longitudinal studies hampers a comprehensive understanding of the chronic and cumulative effects of heat exposure, an aspect of particular importance given the progressive nature of many heat-related conditions, including renal and cardiovascular diseases [16]. Also notable is the limited attention to vulnerable subpopulations: women [1, 16, 20], migrant workers [1, 3, 5, 16, 19], indoor workers, particularly those operating in poorly ventilated or inadequately cooled settings [2, 6, 15, 16], or residents of low- and middle-income countries [3, 16]. This lack is not insignificant, as these groups represent a growing share of the global workforce and may be disproportionately exposed to the combined effects of social vulnerability, job insecurity, and reduced access to protection and adaptation measures. Another limitation is the poor attention to less frequently studied outcomes, such as mental health [17, 19], productivity [5, 14], and chronic diseases [16] other than heatstroke. The tendency to focus on acute and immediately observable events (accidents, heat strain) [6] may lead to underestimating long-term effects and those that are harder to measure, yet equally relevant for public health and production systems sustainability. From this perspective, emerging outcomes (mental health, kidney disease, reduced productivity) should not be considered incidental, but integral to modern occupational risk assessment.

A key methodological challenge in occupational epidemiology that deserves further attention is the attribution of health effects specifically to occupational heat exposure, as opposed to general environmental exposure. In many contexts, particularly in urban settings, workers are simultaneously exposed to ambient heat affecting the general population. This creates a strong collinearity between occupational and environmental exposures, making it difficult to distinguish their respective contributions. In this context, inadequate control of confounding factors (such as socio-economic status, baseline health conditions, urban heat island effects, and co-exposures) may bias risk estimates. Importantly, this bias is likely to act in the direction of overestimating occupational risks, as part of the observed associations may reflect broader environmental exposure rather than strictly work-related conditions. The magnitude of this potential overestimation is difficult to quantify based on the available evidence, but it represents a critical limitation that should be considered when interpreting the findings. Future studies should prioritize study designs and exposure assessment strategies capable of better distinguishing occupational from non-occupational heat exposure.

Generally, the preventive interventions and mitigation strategies identified in the reviews (Table S4) confirm that workplace adaptation to climate change requires multi-level approaches, ranging from individual measures to organizational, engineering, and policy interventions [1]. However, the evidence supporting the effectiveness of these interventions remains largely descriptive, with limited quantitative evaluation and substantial variability across study designs and contexts.

At the organizational/managerial level, measures such as shift reorganization, scheduled breaks, and worker training are effective, but frequently implemented inconsistently, particularly in small enterprises or low-regulation contexts [1, 18]. At the engineering/environmental level, ventilation, shading, cooling, and access to safe drinking water represent concrete environmental controls, even if their implementation is limited by logistical and economical constraints, especially in agriculture and low-income countries [1, 15-16, 20]. Individual strategies (hydration, technical clothing, self-recognition of symptoms) are crucial but raise the issue of worker empowerment and responsibility: while self-regulation can be effective, it risks shifting responsibility from employer to worker, especially in precarious or poorly trained working conditions [1, 16]. Regarding regulatory frameworks and preventive governance tools, there is a clear need for binding standards rather than voluntary recommendations. Some countries have introduced local exposure thresholds and work-specific heat-alert systems, but their deployment remains fragmented [6, 18]. Finally, training and communication are also crucial [1, 18] but translating awareness into sustained behavioral change remains challenging. More specifically, the three additional studies focusing on preventive interventions against heat stress [11-13] reinforce the need for a multifactorial and integrated approach to occupational heat prevention. These reviews highlight that effective management of heat-related risks in the workplace requires the simultaneous implementation of internal (physiological and behavioral), external (technical and environmental), and organizational (administrative and managerial) interventions. At the engineering and environmental level, the evidence confirms the efficacy of cooling technologies in reducing core body temperature and improving thermal comfort. These interventions, already tested in sectors such as construction and heavy industry, demonstrate moderate to high effectiveness, especially when combined with ventilation, shading, or misting systems. However, as underlined by the reviewed studies, their large-scale application remains limited by costs, maintenance demands, and contextual feasibility, particularly in agriculture and in low-resource settings. Even apparently simple environmental measures, such as access to shaded areas or safe drinking water, may encounter logistical barriers that make them difficult to sustain over time. At the organizational level, the implementation of work-rest cycles, acclimatization plans, and training programs emerges as a key element of prevention. Structured schedules allowing recovery in cool areas, along with health and safety education and awareness initiatives, have proven to reduce the incidence of heat-related illness. Nevertheless, their effectiveness depends strongly on managerial commitment and compliance monitoring. The introduction of continuous monitoring systems, including wearable sensors capable of tracking physiological indicators in real time, is proposed as a promising innovation to support data-driven prevention, although further field validation is needed. At the individual level, strategies such as adequate hydration, appropriate clothing, and self-recognition of heat stress symptoms play a central role. The use of hydration backpacks, electrolyte replacement solutions, and cooling ingestion techniques (e.g., ice slurry) has been shown to effectively improve physiological recovery. However, these approaches should complement and not replace structural interventions. As said, collectively, these findings emphasize that organizational and behavioral measures alone are insufficient unless supported by technical and environmental improvements and by institutional governance frameworks. Accordingly, preventive interventions and mitigation strategies must be therefore tailored to different production and territorial contexts, balancing technical feasibility, economic sustainability, and social acceptability. The evidence on preventive interventions remains limited and heterogeneous. Differences in intervention types, outcome definitions, and study populations limit comparability, while the concentration of evidence in specific sectors and settings limits generalisability. Further research with more standardised approaches is needed to strengthen the evidence base. From this perspective, it becomes clear that adaptation to climate change cannot be entrusted to individual actors but requires coordinated, multi-level governance, in which public policies, corporate regulations, and individual responsibilities are coherently aligned [2, 16, 25].

Conclusion

The evidence summarized in this umbrella review confirms that climate change is emerging as a determinant of workers' health and safety. The most consistent findings concern outdoor occupations, especially in agriculture and construction, where heat exposure is linked to a higher likelihood of injuries and heat strain. At the same time, increasing attention to other outcomes, including mental health, renal disease, and chronic conditions, reveals how climate-related risks extend beyond acute heat effects only. From a methodological perspective, the available literature remains fragmented: narrative reviews still predominate, while systematic approaches and meta-analyses are relatively uncommon. This fragmentation partly reflects the challenges of integrating diverse study designs, exposure measures, and health endpoints. In addition to these research gaps, the literature documents a variety of preventive strategies. However, their implementation in real work settings remains inconsistent and is rarely subjected to rigorous evaluation, limiting their potential impact on workers' protection. From an ethical and regulatory standpoint, it is important to emphasize that individual-level strategies should complement, and never replace, structural and organizational interventions. Placing excessive reliance on individual measures risks shifting the primary responsibility for worker protection from the employer to the worker, which is both ethically problematic and potentially incompatible with occupational health and safety legislation in many jurisdictions. Ensuring workers' health and safety in the context of climate change remains, first and foremost, an institutional and employer responsibility. Future studies should aim to improve methodological comparability, adopt longitudinal designs, and focus more on underrepresented populations. Bridging research and practice will require close collaboration between disciplines such as climatology, epidemiology, occupational hygiene, ergonomics, and policy studies.

In conclusion, environmental heat exposure at work should be considered a major occupational health concern with direct implications for workers' safety and health. Progress will depend on evidence-based prevention, coordinated research, and policy efforts that translate scientific knowledge into practical prevention.

Supplementary Materials: The following are available online: Figure S1: Evaluation of risk of bias in included reviews using ROBIS tool. D1: Domain - Study eligibility criteria; D2: Domain - Identification and selection of studies; D3: Domain - Data collection and study appraisal; D4: Domain - Synthesis and findings; Table S1: Research query used in this umbrella review; Table S2: Results in terms of study population and main identified health outcomes by review type; Table S3: Limitations, future developments and conclusions outcomes from the considered reviews by review type; Table S4: Policy recommendation and measures to reduce heat stress by review type; Table S5: Characteristics of reviews on intervention to prevent heat-related health and safety effects; Table S6: Principal results of reviews on intervention to prevent heat-related health and safety effects

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